Section 3.1 IM-OLS Cointegration Estimation Matrix: U.S. NFC Transformation Elasticity (1929–2024)

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Data Provenance and Estimation Summary

This report presents the empirical cointegration results for Section 3.1 of Chapter 2. All models are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over the full NIPA sample window (1931–2024, $T = 94$ observations after listwise deletion for the lag-1 conditioning variables). To resolve multi-collinearity, all interactions are orthogonalized via Frisch-Waugh-Lovell (FWL) against their base regressors and any detected Bai-Perron step dummies.

A Cointegration Diagnostics and Asymptotic Null Test Behavior. Standard residual-based tests for cointegration in non-linear polynomial regressions require careful treatment of deterministic drifts and partial sum processes. In our estimation triad, the Shin (1994) LM stationarity test (C_{Shin}) evaluates the null hypothesis of cointegration (H_0: Cointegration) directly on the Integrated Modified OLS residuals $\hat{u}_t^+$. Because the IM-OLS estimator is intrinsically specified in partial-sum space, the fitted residuals $\hat{u}_t^+$ already represent partial sums $S_t = \sum_{i=1}^{t} e_i$; evaluating the test without re-accumulating residuals avoids an artificial $I(2)$ divergence. Concurrently, the Vogelsang–Wagner (2014) Type 2 Wald test ($S_{\text{Type 2}}$) detrends the integrated capital regressors x_t against the deterministic trend matrix $\mathbf{S}_d$ prior to constructing the reverse cumulative sum regressors Z_{2t}. This prevents unspanned polynomial drifts from blowing up the test statistic. Across both Specification Set 1 (double interaction) and Specification Set 2 (triple effective wage interaction), the residual ADF tests reject the null of no cointegration at standard significance thresholds once structural break step shifts ($\mathbf{D}_t^{\text{step}}$) are incorporated, while the Shin LM and VW Type 2 tests fail to reject the null of cointegration ($C_{\text{Shin}} < 0.121$, $S_{\text{Type 2}} < 13.78$), establishing robust cointegration across all preferred specifications.

Table 1: Model Configuration Cross-Walk

Model ID	Spec Set	Capital Measure (k_t)	Wage Share Measure (ω_{t-1})	Structural Breaks
M1.1a	Set 1	Private Core (ME+NRC)	Baseline NFC NVA	No
M1.1b	Set 1	Private Core (ME+NRC)	Baseline NFC NVA	Yes
M1.2a	Set 1	Infra-Augmented (+K_GT)	Baseline NFC NVA	No
M1.2b	Set 1	Infra-Augmented (+K_GT)	Baseline NFC NVA	Yes
M1.3a	Set 1	Private Core (ME+NRC)	Corporate NVA (Finance-Adjusted)	No
M1.3b	Set 1	Private Core (ME+NRC)	Corporate NVA (Finance-Adjusted)	Yes
M1.4a	Set 1	Infra-Augmented (+K_GT)	Corporate NVA (Finance-Adjusted)	No
M1.4b	Set 1	Infra-Augmented (+K_GT)	Corporate NVA (Finance-Adjusted)	Yes
M1.5a	Set 1	Private Core (ME+NRC)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	No
M1.5b	Set 1	Private Core (ME+NRC)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	Yes
M1.6a	Set 1	Infra-Augmented (+K_GT)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	No
M1.6b	Set 1	Infra-Augmented (+K_GT)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	Yes
M2.1a	Set 2	Private Core (ME+NRC)	Baseline NFC NVA	No
M2.1b	Set 2	Private Core (ME+NRC)	Baseline NFC NVA	Yes
M2.2a	Set 2	Infra-Augmented (+K_GT)	Baseline NFC NVA	No
M2.2b	Set 2	Infra-Augmented (+K_GT)	Baseline NFC NVA	Yes
M2.3a	Set 2	Private Core (ME+NRC)	Corporate NVA (Finance-Adjusted)	No
M2.3b	Set 2	Private Core (ME+NRC)	Corporate NVA (Finance-Adjusted)	Yes
M2.4a	Set 2	Infra-Augmented (+K_GT)	Corporate NVA (Finance-Adjusted)	No
M2.4b	Set 2	Infra-Augmented (+K_GT)	Corporate NVA (Finance-Adjusted)	Yes
M2.5a	Set 2	Private Core (ME+NRC)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	No
M2.5b	Set 2	Private Core (ME+NRC)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	Yes
M2.6a	Set 2	Infra-Augmented (+K_GT)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	No
M2.6b	Set 2	Infra-Augmented (+K_GT)	Shaikh-Tonak Tier 2 (Supervisory-Corrected)	Yes

1 Specification Set 1: Double Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1}$)

Table 2: Specification Set 1: Double Interaction FWL-Orthogonalized IM-OLS

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)												
Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
$\hat{\alpha}$ (Intercept)	6.57***	6.03***	6.47***	6.30***	6.56***	5.93***	6.47***	6.20***	6.34***	5.05***	6.19***	4.44***
	(0.32)	(0.73)	(0.33)	(0.68)	(0.32)	(0.76)	(0.33)	(0.71)	(0.22)	(0.75)	(0.23)	(0.68)
$\hat{\beta}$ (Private Core k_{priv})	0.58***	0.60***	—	—	0.58***	0.61***	—	—	0.60***	0.70***	—	—
	(0.02)	(0.06)			(0.02)	(0.07)			(0.02)	(0.06)		
$\hat{\beta}$ (Infra-Augmented k_{aug})	—	—	0.58***	0.57***	—	—	0.58***	0.57***	—	—	0.59***	0.74***
			(0.02)	(0.06)			(0.02)	(0.06)			(0.02)	(0.05)
$\hat{\theta}$ (Orthogonalized Interaction)	-0.02	-0.08	0.02	-0.07	-0.03	-0.04	0.01	-0.03	-0.24	-0.27	-0.24	-0.27
	(0.13)	(0.10)	(0.13)	(0.09)	(0.12)	(0.10)	(0.12)	(0.09)	(0.09)	(0.12)	(0.09)	(0.10)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.41	-2.81	-2.22	-2.79	-2.33	-2.73	-2.16	-2.73	-2.94	-2.95	-2.77	-3.50*
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.04	0.04	0.03	0.04	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	3.74	10.93	2.70	11.41	3.22	10.94	0.05	11.94	0.11	10.70
Bandwidth Ratio (b)	0.2746	0.1872	0.3037	0.1844	0.2864	0.1934	0.3145	0.1892	0.1940	0.1935	0.2037	0.1478
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Stars for coefficient rows and the Residual VW Type 2 cointegration test are derived from a custom Monte Carlo simulation suite mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, applying FWL orthogonalization in simulation). Distributional variables centered at 1929 initial state (t_0).

2 Specification Set 2: Triple Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1} \cdot (1 - \iota_{t-1})$)

Table 3: Specification Set 2: Triple Interaction FWL-Orthogonalized IM-OLS

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)												
Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
$\hat{\alpha}$ (Intercept)	6.59***	6.47***	6.48***	6.74***	6.58***	6.39***	6.48***	6.67***	6.30***	5.68***	6.16***	5.34***
	(0.32)	(0.64)	(0.33)	(0.60)	(0.32)	(0.66)	(0.33)	(0.62)	(0.20)	(0.34)	(0.21)	(0.35)
$\hat{\beta}$ (Private Core k_{priv})	0.58***	0.56***	—	—	0.58***	0.57***	—	—	0.60***	0.65***	—	—
	(0.02)	(0.06)			(0.02)	(0.06)			(0.01)	(0.03)		
$\hat{\beta}$ (Infra-Augmented k_{aug})	—	—	0.58***	0.52***	—	—	0.58***	0.53***	—	—	0.60***	0.66***
			(0.02)	(0.05)			(0.02)	(0.05)			(0.01)	(0.03)
$\hat{\theta}$ (Orthogonalized Interaction)	0.00	-0.17	0.03	-0.16	-0.00	-0.15	0.02	-0.15	-0.29*	-0.34**	-0.27*	-0.33*
	(0.10)	(0.07)	(0.10)	(0.07)	(0.10)	(0.07)	(0.10)	(0.07)	(0.08)	(0.08)	(0.09)	(0.08)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.32	-3.06	-2.17	-3.02	-2.26	-2.96	-2.12	-2.95	-2.96	-3.82**	-2.72	-3.78**
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.05	0.04	0.06	0.04	0.05	0.04	0.04	0.03	0.04	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	8.48	9.03	6.17	9.61	6.27	9.33	0.46	13.69	0.78	12.58
Bandwidth Ratio (b)	0.2976	0.1687	0.3213	0.1632	0.3066	0.1732	0.3291	0.1675	0.1873	0.1651	0.2024	0.1418
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Stars for coefficient rows and the Residual VW Type 2 cointegration test are derived from a custom Monte Carlo simulation suite mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, applying FWL orthogonalization in simulation). Distributional variables centered at 1929 initial state (t_0).